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United States Patent	12405515
Kind Code	B2
Date of Patent	September 02, 2025
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Electronic device

Abstract

An imaging device is configured to include a finder and an eyecup unit. An eyecup is provided so as to surround an eyepiece window formed on a cover member. A base member holds the eyecup and the cover member. The eyecup is configured to include a curved and hollow eyepiece region, a first holding region comprising a first hole group, and a second holding region comprising a second hole group. The base member is configured to include a first protrusion group corresponding to the first hole group and a second protrusion group corresponding to the second hole group. The first hole group fits with the first projection group, and the second hole group fits with the second projection group.

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Appl. No.: **17/839686**

Filed: **June 14, 2022**

Prior Publication Data

Document Identifier	Publication Date
US 20220413363 A1	Dec. 29, 2022

Foreign Application Priority Data

JP	2021-107986	Jun. 29, 2021

Publication Classification

Int. Cl.: **G03B13/06** (20210101)

U.S. Cl.:

CPC **G03B13/06** (20130101);

Field of Classification Search

CPC: G02B (23/2476); G02B (23/16); G03B (11/04); G03B (13/06); G03B (11/046)

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Background/Summary

BACKGROUND OF THE INVENTION
Field of the Invention
(1) The present invention relates to an electronic device that includes an eyepiece member in a finder.
Description of the Related Art
(2) A finder provided in an imaging device includes a configuration for a photographer to observe an image of a subject guided to an imaging sensor. A subject field angle, setting information, a photographed image and the like are displayed in the finder. The photographer is able to look into the finder in an eyepiece state and confirm the display content. An eyecup provided in an eyepiece unit of the finder is a member configured to block external light and make it easy for a photographer to look into the eyepiece unit. An eyecup is often formed of an elastic member, and is easily deformed when pulled. Therefore, it is necessary to take measures against the eyecup

becoming detached from an exterior portion or, if the eyecup is used in a deformed state, its shape not fitting the shape of the eye periphery.
(3) Japanese Patent Publication Laid-Open No. 2016-53658 discloses a viewfinder provided with a hollow cover attached to an imaging device body and an eyecup attached to the outside surface of the cover. The cover is divided in two, consisting of an upper cover and a lower cover. A screw is provided in a space for disposing a fixing screw for fixing the upper cover and the lower cover, and a hole through which the fixing screw is passed and a groove shape for passing a screwdriver are provided. A protruding shape (pulling prevention rib) formed on the inner surface of the eye cup is fitted with a groove shape for passing the screwdriver.
(4) In the conventional technology disclosed in Japanese Patent Publication Laid-Open No. 2016-53658, because a groove shape for passing a fixing screw is provided in the optical axis direction of the eyecup, it is necessary to secure the size of the eyecup in the optical axis direction. Accordingly, this may be a cause of inhibiting space saving.
SUMMARY OF THE INVENTION
(5) The present invention provides an electronic device capable of fixing a component of an eyepiece member without using an adhesive and the like, capable of saving space, and capable of suppressing detachment of a component that configures the eyepiece member.
(6) An electronic device according to an embodiment of the present invention is an electronic device comprising an eyepiece member in a finder provided on a main body, the eyepiece member comprising a first member in which an eyepiece window of the finder is formed, a second member provided around the eyepiece window, and a third member configured to hold the first and second members, wherein, in the second member, an eyepiece region is hollow, and a first and a second hole group is included, wherein the third member includes a first protrusion group that protrudes towards a side of the main body and fits with the first hole group, and the first or the third member includes a second protrusion group that fits with the second hole group.
(7) Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIGS. **1**A and **1**B are external perspective views of an imaging device according to an embodiment of the present invention.
(2) FIG. **2** is a block diagram showing a functional configuration of an imaging device according to an embodiment of the present invention.
(3) FIGS. **3**A and **3**B are exploded perspective views of an eyecup unit according to a first embodiment of the present invention.
(4) FIGS. **4**A to **4**E are diagrams showing an eyecup unit according to a first embodiment of the present invention.
(5) FIGS. **5**A and **5**B are diagrams showing an eyecup unit according to a second embodiment of the present invention.
DESCRIPTION OF THE EMBODIMENTS
(6) Hereinafter, preferred embodiments of the present invention will be explained in detail with reference to the accompanying drawings. The present invention can be applied to various electronic devices that includes an eyepiece member in a finder. In the embodiments, an example of an imaging device configured to include an eyecup in a finder eyepiece unit is shown as an electronic device.
First Embodiment
(7) Referring to FIG. **1**A to FIG. **2**, a digital camera configured to include an interchangeable lens

(hereinafter, simply referred to as a "camera") **10** will be explained as an imaging device according to the present embodiment. FIGS. **1**A and **1**B show a main body unit **100** in a state in which an interchangeable lens unit is not attached. The positional relationship of each unit is explained by defining the subject side as the front surface side. FIG. **1**A is a perspective view when a camera **10** is viewed from the front surface side, and FIG. **1**B is a perspective view when the camera **10** is viewed from the back surface side. The optical axis direction of the camera **10** is defined as the Z direction, and the two directions orthogonal to the Z direction are defined as the X direction and the Y direction, respectively. The horizontal direction is defined as the X direction, and the vertical direction is defined as the Y direction. FIG. **2** is a block diagram showing the functional configuration of the camera **10**.

(8) A grip unit **120** is a holding unit for a user to hold the camera **10**. The grip unit **120** is provided at one end of the main body unit **100**, and is formed in a curved shape so as to be grasped by the user by covering with the palm and hooking the fingers. By forming the surface of the grip unit **120** with an elastic member such as synthetic rubber, it is possible to obtain good gripping performance.

(9) The camera **10** is configured so that a lens unit **50** (refer to FIG. **2**) can be attached and detached with respect to a mount opening unit **60** of the front surface by a pressing operation of a lens release button **61**. The lens unit **50** configures an imaging optical system, and is configured to include a focus lens **51***a* and a zoom lens **51***b* consisting of a plurality of lens groups. The lens unit **50** is configured to include an aperture **52** that adjusts an opening amount. A lens drive mechanism **53** drives the focus lens Ma and the zoom lens **51***b*, and performs focusing and zoom driving. An aperture drive mechanism **54** drives the aperture **52**, and controls an aperture value. A lens CPU (central processing unit) **55** performs various signal processing, and controls each unit in the lens unit **50**.

(10) The main body unit **100** and the lens unit **50** are electrically connected by a camera-side communication I/F (interface) unit **62** and a lens-side communication I/F unit **56**, and can communicate with each other. In addition, power supply from the main body unit **100** to the lens unit **50** is performed.

(11) A camera CPU **40** is a central unit that controls an imaging system, and performs operation control of each element of the camera **10**. Hereinafter, the camera CPU **40** is simply referred to as "CPU **40**". A lens detection switch **63** of the main body unit **100** outputs a detection signal, which determines whether the main body unit **100** and the lens unit **50** can communicate via the camera-side communication I/F unit **62** and the lens-side communication IF unit **56**, to the CPU **40**. In addition, the lens detection switch **63** outputs a signal for identifying the type of lens unit **50** mounted on the main body unit **100** to the CPU **40**.

(12) A power supply **65** provides power to each element of the camera **10**. The power supply **65** is configured by a battery pack and the like attachable to and detachable from the main body unit **100**. A power supply circuit **66** converts the voltage of the power supply **65** into a voltage required for the operation of each element of the camera **10**, and supplies the voltage to a circuit unit.

(13) A shutter **80** consists of a focal plane shutter and is driven by a shutter drive circuit **81**, and performs control of the incidence of an imaging light flux by exposure and shielding of an imaging sensor **71**. The shutter drive circuit **81** can move to and hold an exposed state (open state) or a shielded state (closed state) of the imaging sensor **71** by opening and closing a shutter curtain (not shown).

(14) The imaging sensor **71** captures the imaging light flux from the lens unit **50** and performs photoelectric conversion. For example, a CMOS (complementary metal oxide semiconductor) imaging sensor or a CCD (charge-coupled device) imaging sensor is used as the imaging sensor **71**, and includes an electronic shutter function.

(15) An optical low-pass filter **75** disposed on the front surface side of the imaging sensor **71** is a rectangular optical element consisting of a material such as quartz crystal. A piezoelectric element **76** is adhesively attached to the surface of the optical low-pass filter **75**, and a piezoelectric element

drive circuit **77** performs energization of the piezoelectric element **76**. The piezoelectric element drive circuit **77** is electrically connected to the piezoelectric element **76** via a piezoelectric element flexible substrate (not shown). By vibrating the optical low-pass filter **75** in the Z direction in a plurality of vibration modes of different orders through the energization control of the piezoelectric element **76**, it is possible to remove dust particles adhering to the surface of the optical low-pass filter **75**.

(16) A shake detection sensor **78** is, for example, an angular velocity sensor, and periodically detects the angular velocity of the shake of the camera **10** and converts the angular velocity into an electrical signal. The output of the shake detection sensor **78** is acquired by the CPU **40** as a detection signal of the shake amount of the camera **10**.

(17) An imaging unit **70** includes an imaging sensor **71**, an optical low-pass filter **75**, a piezoelectric element **76**, and imaging unit drive mechanism **72**. An imaging unit drive mechanism **72** is a mechanism for driving the imaging unit **71** within a predetermined plane, for example, a drive coil, a permanent magnet, and a position detection sensor (all not shown) are provided to drive the imaging sensor **71** on a plane orthogonal to the optical axis of the camera **10**.

(18) An imaging unit drive circuit **73** is electrically connected to an imaging unit drive mechanism **72** via a flexible substrate for the imaging unit drive mechanism (not shown), and performs energization control of the imaging unit drive mechanism **72**. The CPU **40** acquires a detection signal from the shake detection sensor **78**, and performs drive control of the imaging unit **70** in a direction to cancel out the shake of the camera **10** according to the detection result. Thereby, an image blur caused by the shake of the camera **10** can be corrected.

(19) An external memory **67** consists of, for example, a semiconductor memory card attachable to and detachable from the main body unit **100**, and stores data such as a photographed image. An operation member **110** is a member for the user to perform an operation instruction to the camera **10**. The CPU **40** receives an instruction signal from each type of the operation members **110** and performs each type of control. For example, FIG. **1**A shows a shutter button **111** configured to perform an instruction of an imagine operation, and FIG. **1**B shows a selection button **114** for selecting various settings, and a setting button **115** for determining various settings.

(20) The camera **10** is configured to provide a first display **200** and a second display **210**. As shown in FIG. **1**B, the first display **200** is provided on the back surface of the camera **10**. The first display **200** is configured by an LCD (liquid crystal display) or the like, and a finder **85** is provided on the upper side (+Y side) thereof so that the user can observe the subject.

(21) The finder **85** is an electronic viewfinder that includes an eyepiece optical system **86** provided at the upper left portion of the back surface of the camera **10**. The second display **210** configures a display unit of the electronic viewfinder. The finder **85** performs display of a through image by the imaging sensor **71** and a setting display of the camera **10**.

(22) A diopter adjustment dial **87** (refer to FIG. **1**B) is a diopter adjustment member for a user to perform an operation, and is provided on an exterior portion of the main body unit **100**. The diopter adjustment dial **87** is connected to an eyepiece optical system **86**. The eyepiece optical system **86** is configured to advance and retract in the Z direction by the rotation operation of the diopter adjustment dial **87**. The user can perform adjustment of the diopter to suit their own eyes.

(23) An eyecup unit **300** is attached to the finder **85**. There is an embodiment of a configuration in which the eyecup unit **300** is removable from the main body unit **100** of the camera **10**, and a further embodiment of a configuration in which the eyecup unit **300** is fixed to the main body unit **100**.

(24) The finder **85** protrudes from the main body unit **100** toward the back surface side (−Z side). When the user views the main body unit **100** of the camera **10** from the back surface side, the finder **85** is positioned above the first display **200** when the user holds the camera **10** by gripping the grip unit **120** with the right hand.

(25) Next, with reference to FIG. **3**A to FIG. **4**E, the eyecup unit **300** according to the present

embodiment will be described in detail. FIG. **3**A is an exploded perspective view of the camera **10** and eyecup unit **300** when viewed from the back surface side. FIG. **3**B is an exploded perspective view of the eyecup unit **300** when viewed from the front surface side.

(26) The eyecup unit **300** is configured by an eyecup **310**, a base member **320**, and a cover member **330**. The eyecup **310** is a member that comes in contact with the face (the periphery of the eye) when the user looks into the finder **85**, and is formed of a soft material such as silicone or rubber. The base member **320** is a member that holds the eyecup **310**, and is attached or fixed to the finder **85**. The cover member **330** is a member that includes an eyepiece window. The base member **320** and the cover member **330** are formed of hard material such as PC (polycarbonate) resin or ABS (Acrylonitrile Butadiene Styrene) resin.

(27) A hollow eyepiece region **310***a* in the eyecup **310** is a region that is in contact with the face of the photographer when the photographer looks into the finder **85**. The eyecup **310** includes a first holding region **310***b* and a second holding region **310***c*. The first holding region **310***b* is a region that extends towards the outer peripheral side of the eyepiece region **310***a* with respect to the finder **85**, and includes a plurality of hole portions formed therein. Hereinafter, these hole portions are referred to as a “first hole group **310***d*”. The second holding region **310***c* is a region that extends towards the inner peripheral side of the eyepiece region **310***a* with respect to the finder **85**, and includes a plurality of hole portions formed therein. Hereinafter, these hole portions are referred to as a “second hole group **310***e*”.

(28) The base member **320** is formed by a first protrusion group **320***a* and a second protrusion group **320***b*, and an engaged portion **320***c*. The first protrusion group **320***a* is configured by a plurality of protrusions that protrude toward the front surface side (+Z side) and fit into the first hole group **310***d*. The second protrusion group **320***b* is configured by a plurality of protrusions that protrude toward the back surface side (−Z side) and fit into the second hole group **310***e*. The engaging portion **330***b* of the cover member **330**, which will be described hereinafter, is engaged with the engaged portion **320***c*.

(29) An eyepiece window **330***a* configured to allow a photographer to look into the finder **85** is formed in a portion exposed to the exterior by the cover member **330**. The cover member **330** includes an engaging portion **330***b* formed in a position corresponding to the engaged portion **320***c* of the base member **320**. By engaging the engaging portion **330***b* of the cover member **330** and the engaged portion **320***c* of the base member **320**, the eyecup **310** is supported in a state where it is held between the cover member **330** and the base member **320**.

(30) FIGS. **4**A to **4**C are diagrams of the eyecup unit **300** when viewed from different directions. FIG. **4**A is a diagram of the eyecup unit **300** when viewed from the front surface side, FIG. **4**B is a side surface view, and FIG. **4**C is a diagram of the eyecup unit **300** when viewed from the back surface side. FIG. **4**D and FIG. **4**E show a state in which the eyecup unit **300** of FIG. **4**A is attached to the camera **10**. FIG. **4**D is a cross-sectional view taken along the line A-A shown in FIG. **4**A, and FIG. **4**E is a cross-sectional view taken along the line B-B shown in FIG. **4**A.

(31) As shown in FIG. **4**D, the protrusions that configure the first protrusion group **320***a* of the base member **320** are respectively fitted with the holes that configure the first hole group **310***d* of the eyecup **310**. Note that the first protrusion group **320***a* further enters into a groove unit **130** provided in the camera **10**. The first hole group **310***d* (and the first protrusion group **320***a*) is positioned in a location proximate to the eyepiece region **310**A, indicated by a dotted line in the X-Z plane. That is, the first hole group **310***d* of the eyecup **310** is provided at a position included in a range that includes a portion of the eyepiece region **310***a* on a projection plane perpendicular to the optical axis of the finder **85**. In addition, the second protrusion group **320***b* provided on the base member **320** is loosely fitted into the second hole group **310***e* of the eyecup **310**.

(32) As shown in FIG. **4**E, by engaging the engaging portion **330***b* of the cover member **330** with the engaged portion **320***c* of the base member **320**, the cover member **330** is supported by the base member **320**. In a state where the eyecup unit **300** is attached to the main body unit **100**, the

engaging portion **330***b* is in a state of being opposed to the side surface of a regulating portion **131** of the main body unit **100**.
(33) In a state in which the engaged portion **320***c* of the base member **320** and the engaged portion **330***b* of the cover member **330** are engaged with each other, the eyecup **310** is held between the first holding region **310***b* and the second holding region **310***c*, respectively, and the movement thereof is regulated.
(34) As shown in FIG. **4**A, the first hole group **310***d* and the first protrusion group **320***a* are not provided at the upper (+Y side) corners of the eyecup unit **300** (portion “f” and portion “g” as shown by the dotted lines). In addition, at least one protrusion (FIG. **4**A: **320***a***-2**) of the first protrusion group **320***a* is disposed substantially coaxially with the diopter adjustment dial **87**. Thereby, with respect to an external impact on the eyecup **310**, the concentration of stress to one hole portion of the first hole group **310***d* can be mitigated, and the effect of preventing or suppressing the detachment of the eyecup **310** can be obtained.
(35) In the present embodiment, the first protrusion group **320***a* of the base member **320** protruding towards the side of the finder **85** (main body unit side) has the function of an anchor for preventing or suppressing the deformation of the hollow eyepiece region portion in the eyecup **310** towards the outer peripheral side. In addition, the second protrusion group **320***b* of the base member **320** protruding towards the side of the cover member **330** has the function of an anchor for preventing or suppressing the deformation of the hollow eyepiece region portion in the eyecup **310** towards the inner peripheral side.
(36) According to the present embodiment, while enabling space saving, it is possible to realize a configuration in which the eyecup **310**, which is a component that configures the eyepiece member, is not easily detached. In addition, because the first holding region **310***b* and the second holding region **310***c* of the eyecup **310** do not appear on the exterior of the camera **10**, this contributes to improvement of the appearance thereof.
Modifications of the First Embodiment
(37) In the first embodiment, the second protrusion group **320***b* is formed on the base member **320**. However, in a first modification, a group of protrusions corresponding to the second protrusion group **320***b* is formed on the cover member **330**. That is, the protrusion group included in the cover member **330** and the second hole group **310***e* included in the eyecup **310** are fitted with each other.
(38) In addition, in the first embodiment, the second hole group **310***e* is formed in the eyecup **310**. However, in a second modification, the second hole group is formed in the cover member **330**. In this case, a protrusion group to be fitted with the second hole group is formed in the base member **320** or in the eyecup **310**.
(39) In a third modification, a plurality of projection groups corresponding to the first protrusion group **320***a* and the second protrusion group **320***b*, respectively, are formed in the eyecup **310**. A plurality of hole groups corresponding to the first hole group **310***d* and the second hole group **310***e*, respectively, are formed in the base member **320**.
Second Embodiment
(40) With reference to FIGS. **5**A and **5**B, a second embodiment of the present invention will be explained. In the present embodiment, a configuration that enables both good tactile feel and improved reliability is shown with respect to the eyecup unit **300**. Note that the explanations of matters that are similar to those of the first embodiment are omitted, and the shape of the hollow eyepiece region **310***a* in the eyecup unit **300**, which is a point of difference, will be explained in detail.
(41) FIG. **5**A is a diagram of the eyecup unit **300** when viewed from the back surface side. FIG. **5**B is a cross-sectional view taken along the C-C line in FIG. **5**A. In FIG. **5**B, only the shape on one side of the eyecup unit **360** is shown. In the illustrated X direction, the +X side is the side approaching the optical axis of the finder **85**, and the −X side is the side separated from the optical axis of the finder **85**. In addition, in the Z direction, the +Z side is the subject side, and the −Z side

is the photographer's side or the back side.
(42) The eyepiece region **310***a* is a region that includes a portion that the photographer comes in contact with when looking into the finder **85**, and has a shape that is curved toward the −Z side along the way from the +X side towards the −X side. Due to this shape, light shielding properties can be ensured, and by adopting a hollow structure, a good tactile feel can be ensured.
(43) On the inner surface side of the eyepiece region **310***a* (the side that does not appear on the exterior), a plurality of wall thickness change points are provided. The wall thickness change point is a point where the thickness (wall thickness) of a portion that extends from the +X side towards the −X side in the eyecup **310** changes. When viewed from the Y direction, as shown by the circular frame of the dotted line in FIG. **5**B, the first wall thickness change point **401** is provided from the +X side towards the −Z side, and the second wall thickness change point **402** is provided from the −Z side towards the −X side. Further, a third wall thickness change point **403** is provided between the first wall thickness change point **401** and the second wall thickness change point **402**.
(44) The eyecup **310** includes, in order from the +X side, a first wall thickness portion **411**, a second wall thickness portion **412**, a third wall thickness portion **413**, and a fourth wall thickness portion **414**. From the first wall thickness portion **411** on the +X side, the wall thickness portion changed at the first wall thickness change point **401** is the second wall thickness portion **412**. The wall thickness portion changed at the third wall thickness change point **403** with respect to the second wall thickness portion **412** is the third wall thickness portion **413**. The wall thickness portion changed at the second wall thickness change point **402** with respect to the third wall thickness portion **413** is the fourth wall thickness portion **414**. That is, in the present embodiment, three wall thickness change points **401**, **402**, and **403** are provided when viewed from the Y direction, so that the wall thickness can be divided into four different types of wall thickness portions.
(45) These four different types of wall thickness portions will be explained in detail. The thickness of the second wall thickness portion **412** (denoted by “d2”) is set to be equal to or smaller than the thickness of the first wall thickness portion **411** (denoted by “d1”). The thickness of the third wall thickness portion **413** (denoted by “d3”) is set to be smaller than the thickness of the fourth wall thickness portion **414** (denoted by “d4”). That is, in the present embodiment, the following conditions of an inequality are satisfied:
$d1 \geq d2$ Condition 1:
$d3 < d4$ Condition 2:
Thus, a bent portion is formed based on the first wall thickness change point **401** and the second wall thickness change point **402**.
(46) In the present embodiment, the first wall thickness change point **401** and the second wall thickness change point **402** are set so as to be substantially the same height (distance in the Z direction) from the base member **320**, so that the photographing posture is parallel with respect to a subject. Thereby, even in a case in which the user looks into the finder **85** in a state where a predetermined force or greater is applied to the finder **85** at the time the user's eye is in contact with the finder, the hollow eyepiece region **310***a* is deformed so as to be bent with the first wall thickness change point **401** and the second wall thickness change point **402** as fulcrum points.
(47) Further, the base member **320** is configured to include a protruding portion **420** so as to be surrounded in proximity to the hollow eyepiece region **310***a*. The protruding portion **420** is a portion protruding from the base member **320** towards the −Z side. Each of the first wall thickness change point **401** and the second wall thickness change point **402** is provided at a position further toward the back surface side (−Z side) than the back end portion of the protruding portion **420** (the convex portion that protrudes the most towards the back surface side).
(48) With the above configuration, it is possible for a user to maintain a comfortable photographing state without having a sense of incongruity due to contact with the protruding portion **420** at the time of eye contact. The protruding portion **420** plays a role of preventing lateral displacement

from occurring when a force is applied from the +X side towards the −X side with respect to the eyepiece region **310***a*. In a case where there is no possibility of the occurrence of lateral displacement, it is not necessary to provide a protruding portion **420**.
(49) Next, the third wall thickness change point **403** will be explained. The second wall thickness portion **412** and the third wall thickness portion **413** are divided by the third wall thickness change point **403**. In the present embodiment, the following condition of an inequality is satisfied:
d2>d3   Condition 3:
(50) In addition, the second wall thickness portion **412** is extended more towards the −X side than the protruding portion in the Z direction with respect to the rear end portion (the convex portion that protrudes the most towards the back surface side) of the protruding portion **420** provided in the base member **320**. Further, there is no corner at the −X-side tip of the protruding portion **420**, and a rounded shape is formed. For example, it is assumed that the user has dropped the camera **10** from the back side. Even in a case in which the second wall thickness portion **412** is crushed by being in contact with the protruding portion **420**, damage to the eyecup **310** can be prevented or suppressed, and thereby, an eyecup **310** having higher reliability can be realized.
(51) The hollow eyepiece region **310***a* includes a configuration having a shape that is inclined from the +X side towards the −X side, that is, from the inner side towards the outer side. Even when the first wall thickness change point **401** and the second wall thickness change point **402** are provided, the hollow eyepiece region **310***a* is inclined toward the −X side (outer side) at the time of eye contact. The user can comfortably perform photography without having the field of view obstructed.
(52) Above, while the present invention has been described in detail based on the preferred embodiments, these particular embodiments are not intended to limit the present invention, and various modifications in a range that does not depart from the gist of the present invention are included in the scope of the present invention. The embodiments may also be combined as appropriate.
(53) For example, in the embodiments, an example is shown in which the first wall thickness change point **401**, the second wall thickness change point **402**, and the third wall thickness change point **403** are provided on the inner side of the eyepiece region **310***a*, that is, on the side that is not exposed to the exterior. The embodiments are not limited thereby, and even in an embodiment in which the wall thickness change point is provided on the external appearance side of the eyepiece region **310***a*, the effect thereof is similar to that in the present embodiments. The present invention can also be applied to an optical finder.
(54) According to the embodiment described above, it is possible to provide an imaging device mounted with a finder on which an eyecup can be fixed without using adhesives and the like, is effective in saving space, and is more reliable in preventing the eyecup from detaching.
Other Embodiments
(55) While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.
(56) This application claims the benefit of Japanese Patent Application No. 2021-107986, filed Jun. 29, 2021, which is hereby incorporated by reference wherein in its entirety.

Claims

1. An electronic device comprising an eyepiece member in a finder provided on a main body, the eyepiece member comprising: a first member in which an eyepiece window of the finder is formed, a second member provided around the eyepiece window, and a third member configured to hold the first and second members, wherein, in the second member, a portion of an eyepiece region is

hollow, and a first and a second hole group is included, wherein the third member includes a first protrusion group that protrudes towards a side of the main body and fits with the first hole group, wherein the first or the third member includes a second protrusion group that fits with the second hole group, wherein the main body includes a groove, and wherein in a state in which the eyepiece member is attached to the main body, protrusions that configure the first protrusion group and the groove are opposed to each other.
2. The electronic device according to claim 1, wherein the second member includes a first and a second holding region that extends from a portion of the eyepiece region, and wherein a first hole group is formed in the first holding region, and a second hole group is formed in the second holding region.
3. The electronic device according to claim 2, wherein a hole portion that configures the first hole group is not formed at a corner of the first holding region.
4. The electronic device according to claim 1, wherein the first hole group is provided at a position included in a range that includes a portion of the eyepiece region on a projection plane perpendicular to the optical axis of the finder.
5. The electronic device according to claim 1, wherein the finder includes an adjustment member configured to adjust a diopter, and wherein any protrusion in the first protrusion group is disposed substantially coaxially with the adjustment member.
6. The electronic device according to claim 1, wherein in a state in which the eyepiece member is attached to the main body, the first member is located at a position that is separated more from the main body than the third member, and the third member is not exposed to an exterior.
7. The electronic device according to claim 1, wherein the first member includes an engaging portion, wherein the third member includes an engaged portion that corresponds to the engaging portion, and wherein the movement of the second member is regulated by the engagement between the engaging portion and the engaged portion.
8. The electronic device according to claim 7, wherein the main body includes a regulating portion that opposes the engaging portion at a position proximate to the engaging portion.
9. The electronic device according to claim 1, wherein a portion of the eyepiece region has a curved shape, and is configured by a plurality of wall thickness portions having different thicknesses.
10. The electronic device according to claim 9, wherein among a first direction and a second direction orthogonal to an optical axis of the finder, when viewed from the first direction, the plurality of wall thickness portions are divided by a plurality of wall thickness change points, and wherein in a state in which the eyepiece member is attached to the main body, the plurality of wall thickness change points are located at a position that is separated more from the main body than the third member.
11. The electronic device according to claim 10, wherein among the plurality of wall thickness change points, a first wall thickness portion and a second wall thickness portion are divided by a first wall thickness change point, and a third wall thickness portion and fourth wall thickness portion are divided by a second wall thickness change point, and wherein the relationships, the thickness of the first wall thickness portion≥the thickness of the second wall thickness portion, and the thickness of the third wall thickness portion<the thickness of the fourth wall thickness portion, are satisfied.
12. The electronic device according to claim 11, wherein the second first wall thickness portion and the third first wall thickness portion are divided by a third wall thickness change point between the first wall thickness change point and the second wall thickness change point, and wherein the relationship, the thickness of the second wall thickness portion>the thickness of the third wall thickness portion is satisfied.
13. The electronic device according to claim 11, wherein the first wall thickness change point and the second wall thickness change point are set to substantially the same distance with respect to the

third member.
14. The electronic device according to claim 10, wherein the third member is configured to include a protruding portion that protrudes in a third direction away from the main body at a position proximate to a portion of the eyepiece region, and wherein each of the plurality of a wall thickness change points is located at a position that is separated more from the main body than the end portion of the protruding portion.
15. The electronic device according to claim 14, wherein among the first and second wall thickness portions divided by a first wall thickness change point, the second wall thickness portion is extended to a side that is separated more from the first wall thickness portion in a second direction than the end portion of the protruding portion when viewed from the third direction.
16. The electronic device according to claim 10, wherein, when viewed from the first direction, a portion of the eyepiece region is a shape that is inclined with respect to the second direction.
17. The electronic device according to claim 1, further comprising an imaging sensor.
18. An electronic device comprising an eyepiece member in a finder provided on a main body, the eyepiece member comprising: a first member in which an eyepiece window of the finder is formed, a second member provided around the eyepiece window, and a third member configured to hold the first and second members, wherein, in the second member, a portion of an eyepiece region is hollow, and a first hole group is included, wherein the first member includes a second hole group, wherein the third member includes a first protrusion group that protrudes towards a side of the main body and fits with the first hole group, wherein the second or the third member includes a second protrusion group that fits with the second hole group, wherein the main body includes a groove, and wherein in a state in which the eyepiece member is attached to the main body, protrusions that configure the first protrusion group and the groove are opposed to each other.
19. An electronic device comprising an eyepiece member in a finder provided on a main body, the eyepiece member comprising: a first member in which an eyepiece window of the finder is formed, a second member provided around the eyepiece window, and a third member configured to hold the first and second members, wherein, in the second member, a portion of an eyepiece region is hollow, and a first and a second hole group is included, wherein the third member includes a first and a second protrusion group that is fitted with the first and second hole group, respectively, wherein the main body includes a groove, and wherein in a state in which the eyepiece member is attached to the main body, protrusions that configure the first protrusion group and the groove are opposed to each other.